

1.Print all prime numbers between input range (Ex – input 20 50, prints all prime numbers between 20 and 50).

Program

```
start = int(input("Enter starting number: "))
end = int(input("Enter ending number: "))
print("Prime numbers:")
for num in range(start, end + 1):
    if num > 1:
        for i in range(2, num):
            if num % i == 0:
                break
        else:
            print(num, end=" ")
```

Output

OneCompiler Upgrade AI Python Run

main.py × +

```
1 start = int(input("Enter starting number: "))
2 end = int(input("Enter ending number: "))
3 print("Prime numbers:")
4 for num in range(start, end + 1):
5     if num > 1:
6         for i in range(2, num):
7             if num % i == 0:
8                 break
9         else:
10            print(num, end=" ")
```

Console I/O AI Agent

Enter starting number: 20
Enter ending number: 50
Prime numbers:
23 29 31 37 41 43 47

2.Factorial using recursion

Program

```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)
num = int(input("Enter a number: "))
print("Factorial:", factorial(num))
```

Output

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Python

Run

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main.py

1def factorial(n):
2 if n == 0 or n == 1:
3 return 1
4 else:
5 return n * factorial(n - 1)
6num = int(input("Enter a number: "))
7print("Factorial:", factorial(num))

Console

I/O

AI Agent

Enter a number: 5
Factorial: 120

3.Square of numbers using lambda

Program

```
square = lambda x: x * x
num = int(input("Enter a number: "))
print("Square:", square(num))
```

Output

main.py

1square = lambda x: x * x
2num = int(input("Enter a number: "))
3print("Square:", square(num))

Console

I/O

AI Agent

Enter a number: 6
Square: 36

4.Find the second largest element in a list

Program

```
numbers = [10, 25, 8, 45, 32, 18]
numbers = list(set(numbers))
numbers.sort()
print("List:", numbers)
print("Second largest element:", numbers[-2])
```

Output

<pre>main.py × + 1 numbers = [10, 25, 8, 45, 32, 18] 2 numbers = list(set(numbers)) 3 numbers.sort() 4 print("List:", numbers) 5 print("Second largest element:", numbers[-2])</pre>	Console I/O AI Agent <pre>List: [8, 10, 18, 25, 32, 45] Second largest element: 32</pre>
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5.Count frequency of characters in a string

Program

```
text = input("Enter a string: ")
frequency = {}
for char in text:
    if char in frequency:
        frequency[char] += 1
    else:
        frequency[char] = 1
print("Character frequency:")
for char, count in frequency.items():
    print(char, ":", count)
```

Output

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Run

Save

Login

main.py

+

```
1 text = input("Enter a string: ")
2 frequency = {}
3 for char in text:
4     if char in frequency:
5         frequency[char] += 1
6     else:
7         frequency[char] = 1
8 print("Character frequency:")
9 for char, count in frequency.items():
10     print(char, ":", count)
11
```

Console

I/O

AI Agent

4.2s

```
Enter a string: Hi vasavi
Character frequency:
H : 1
i : 2
: 1
v : 2
a : 2
s : 1
```

6.Calculate area of a circle using math library.

Program

```
import math
radius = float(input("Enter radius of the circle: "))
area = math.pi * radius * radius
print("Area of circle:", area)
```

Output

main.py

+

```
1 import math
2 radius = float(input("Enter radius of the circle: "))
3 area = math.pi * radius * radius
4 print("Area of circle:", area)
```

Console

I/O

AI Agent

```
Enter radius of the circle: 13.5
Area of circle: 572.5552611167398
```

7.Reverse a string without using built-in reverse

Program

```
text = input("Enter a string: ")
reversed_text = ""
for char in text:
    reversed_text = char + reversed_text
print("Reversed string:", reversed_text)
```

Output

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AI

Python

Run

main.py

1

text = input("Enter a string: ")

2

reversed_text = ""

3

for char in text:

4

reversed_text = char + reversed_text

5

print("Reversed string:", reversed_text)

Console

I/O

AI Agent

7

Enter a string: Indu
Reversed string: udnI

8.Remove duplicates from a list

Program

```
numbers = [10, 20, 10, 30, 20, 40, 30]
unique_numbers = []
for num in numbers:
    if num not in unique_numbers:
        unique_numbers.append(num)
print("Original list:", numbers)
print("List after removing duplicates:", unique_numbers)
```

Output

OneCompiler

Upgrade

AI

Python

Run

main.py

1

2

3

4

5

6

7

numbers = [10, 20, 10, 30, 20, 40, 30]

unique_numbers = []

for num in numbers:

if num not in unique_numbers:

unique_numbers.append(num)

print("Original list:", numbers)

print("List after removing duplicates:", unique_numbers)

Console

I/O

AI Agent

140 ms

Original list: [10, 20, 10, 30, 20, 40, 30]

List after removing duplicates: [10, 20, 30, 40]

9.Merge two dictionaries

Program

```
dict1 = {"Name": "Indu", "Age": 21}
dict2 = {"Course": "AI&DS", "College": "Saveetha"}
merged_dict = **dict1, **dict2
print("Merged dictionary:")
print(merged_dict)
```

Output

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AI

Python

Run

main.py

1

2

3

4

5

dict1 = {"Name": "Indu", "Age": 21}

dict2 = {"Course": "AI&DS", "College": "Saveetha"}

merged_dict = **dict1, **dict2

print("Merged dictionary:")

print(merged_dict)

Console

I/O

AI Agent

129 m

Merged dictionary:

{'Name': 'Indu', 'Age': 21, 'Course': 'AI&DS', 'College': 'Saveetha'}

10.Fibonacci series using recursion

Program

```
def fibonacci(n):
    if n <= 1:
        return n
    else:
```

```
        return fibonacci(n - 1) + fibonacci(n - 2)
num = int(input("Enter number of terms: "))
print("Fibonacci series:")
for i in range(num):
    print(fibonacci(i), end=" ")
```

Output

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AI Python Run

Save Login

main.py

2.4s

```
1 def fibonacci(n):
2     if n <= 1:
3         return n
4     else:
5         return fibonacci(n - 1) + fibonacci(n - 2)
6 num = int(input("Enter number of terms: "))
7 print("Fibonacci series:")
8 for i in range(num):
9     print(fibonacci(i), end=" ")
```

Console I/O AI Agent

Enter number of terms: 21
Fibonacci series:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181 6765